Computer networking devices

- **Computer networking devices** are units that mediate data in a computer network.
- Computer networking devices are also called network equipment, Intermediate Systems (IS) or InterWorking Unit (IWU).
- Units which are the last receiver or generate data are called hosts or data terminal equipment.

List of computer networking devices

- Common basic networking devices:
  - Repeater
  - Bridge
  - Hub
  - Switch
  - Router
- Some hybrid network devices:
  - Multilayer Switch
  - Bridge Router(Brouter)
  - Digital media receiver

List of computer networking devices

- Hardware or software components that typically sit on the connection point of different networks, e.g. between an internal network and an external network:
  - Proxy
  - Firewall
  - Network Address Translator (NAT)

List of computer networking devices

- Other hardware for establishing networks or dial-up connections:
- Multiplexer
- Network Card
- Modem
- ISDN terminal adapter (TA)
- Line Driver:

Common basic networking devices: Repeater

- Network **repeaters** regenerate incoming electrical, wireless or optical signals.
- Repeaters repeat signals
  - Clean and boost digital transmission
  - Analog networks use amplifiers to boost signal
- Repeaters only work with the physical signal
  - Cannot reformat, resize, or manipulate the data
- Physical layer (layer 1) device

Repeaters (continued)

Figure 2-1 Repeater

Repeaters (continued)

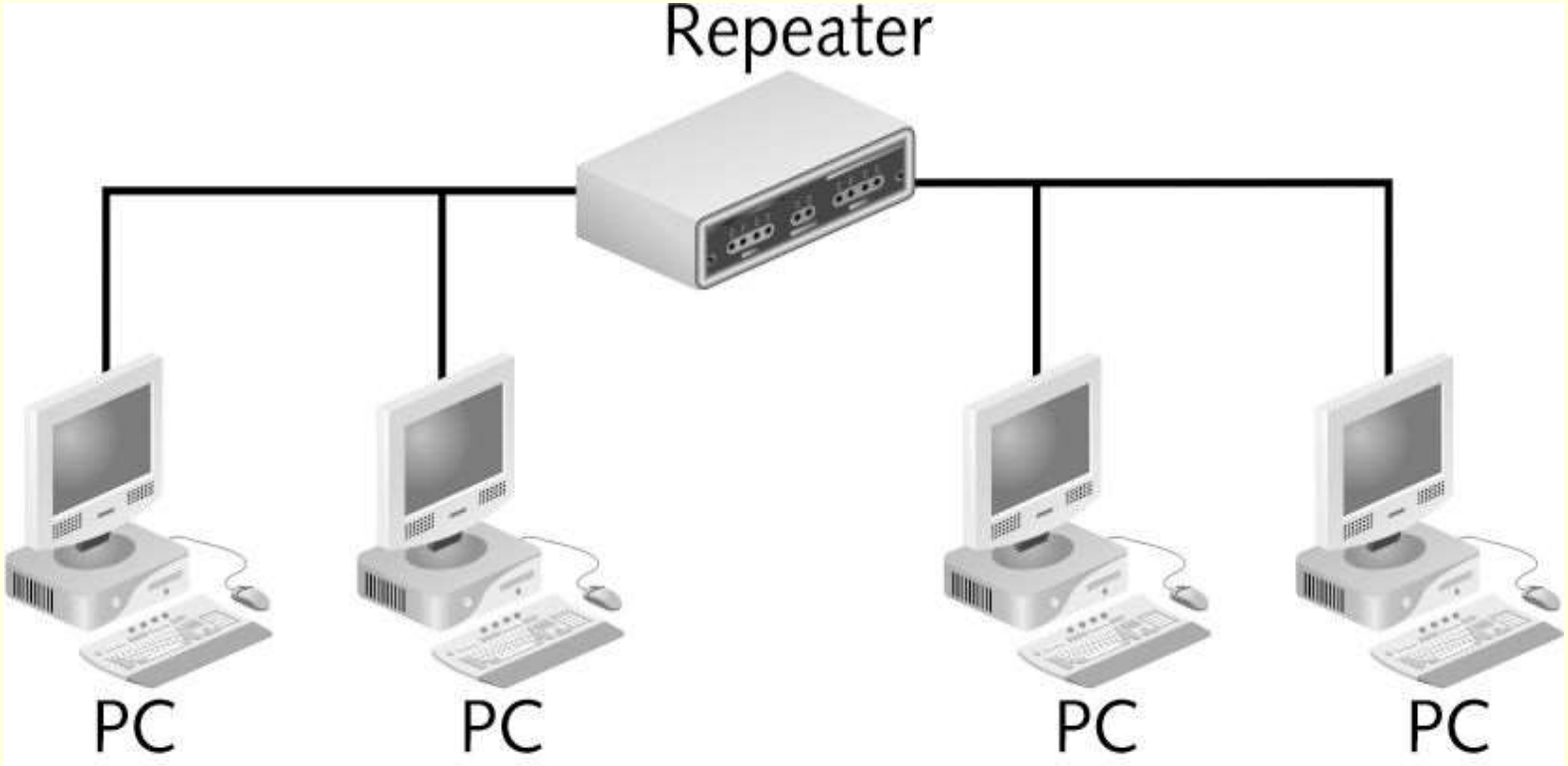


Figure 2-2 Repeater in the network

Advantages of using repeaters

- Extend network physical distance
- Do not seriously affect network performance
- Special repeaters connect different media
  - Copper to fiber

Disadvantages of using repeaters

- Cannot connect different network architectures
  - Token Ring and Ethernet
- Cannot reduce network traffic
- Do not segment the network
  - Repeat everything without discrimination
  - Number of repeaters must be limited
- Repeaters are part of a collision domain

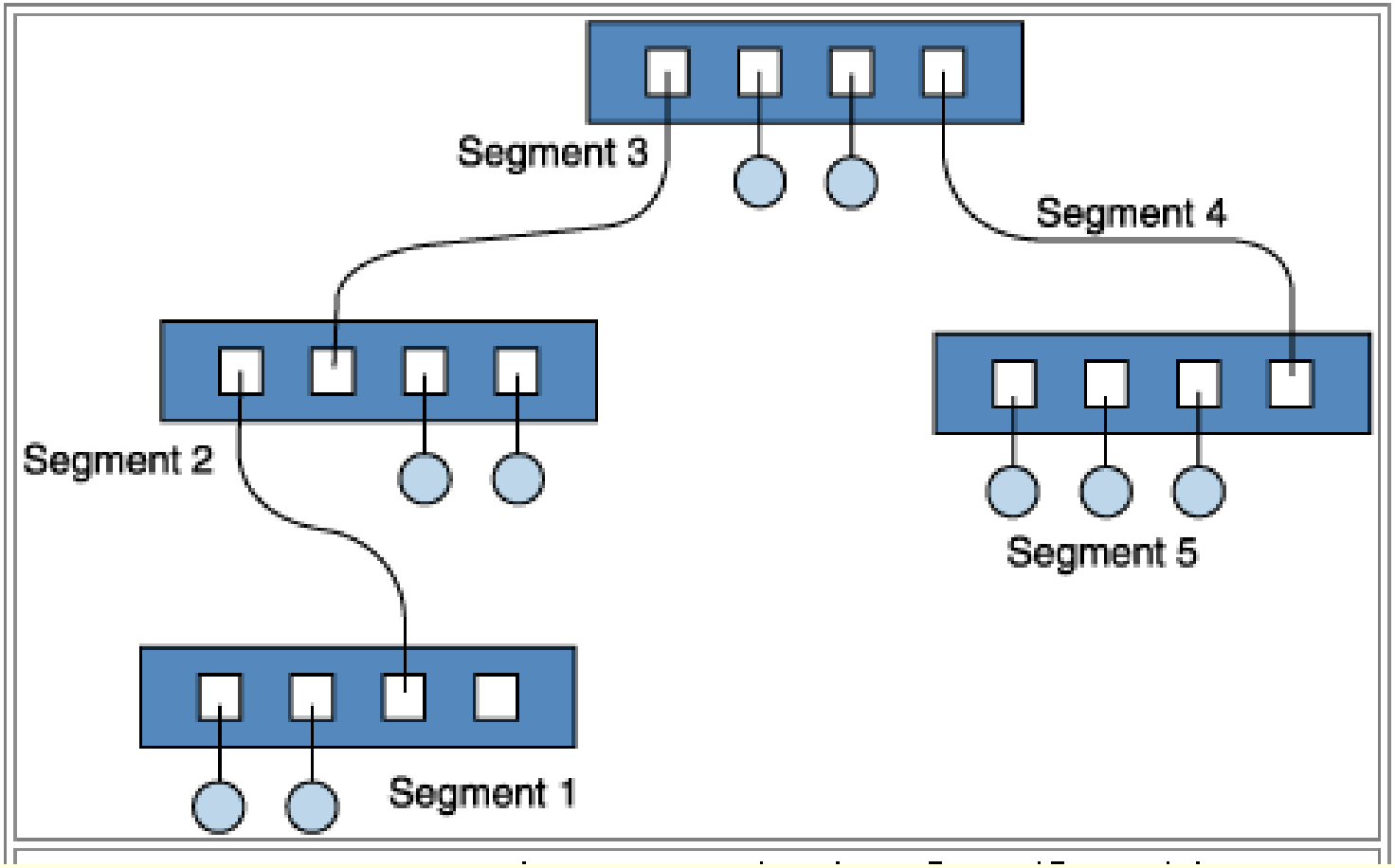
Common basic networking devices: Hub

- A hub joins multiple computers (or other network devices) together to form a single network segment.
- On this network segment, all computers can communicate directly with each other.
- A hub includes a series of ports that each accept a network cable .

Hub

- **Passive hubs** do not amplify the electrical signal of incoming packets before broadcasting them out to the network.
- **Active hubs**, on the other hand, do perform this amplification, as does a different type of dedicated network device called a repeater.
- Some people use the terms **concentrator** when referring to a passive hub and **multiport repeater** when referring to an active hub.

Hubs connecting segments



Hub

- Hubs are classified as Layer 1 (physical layer) devices in the OSI model.
- At the physical layer, hubs support little in the way of sophisticated networking. Hubs do not read any of the data passing through them and are not aware of their source or destination.
- A hub simply receives incoming Ethernet frames, regenerates the electrical signal, and broadcasts these packets out to all other devices on the network.

Hubs (continued)

4-port Ethernet hub

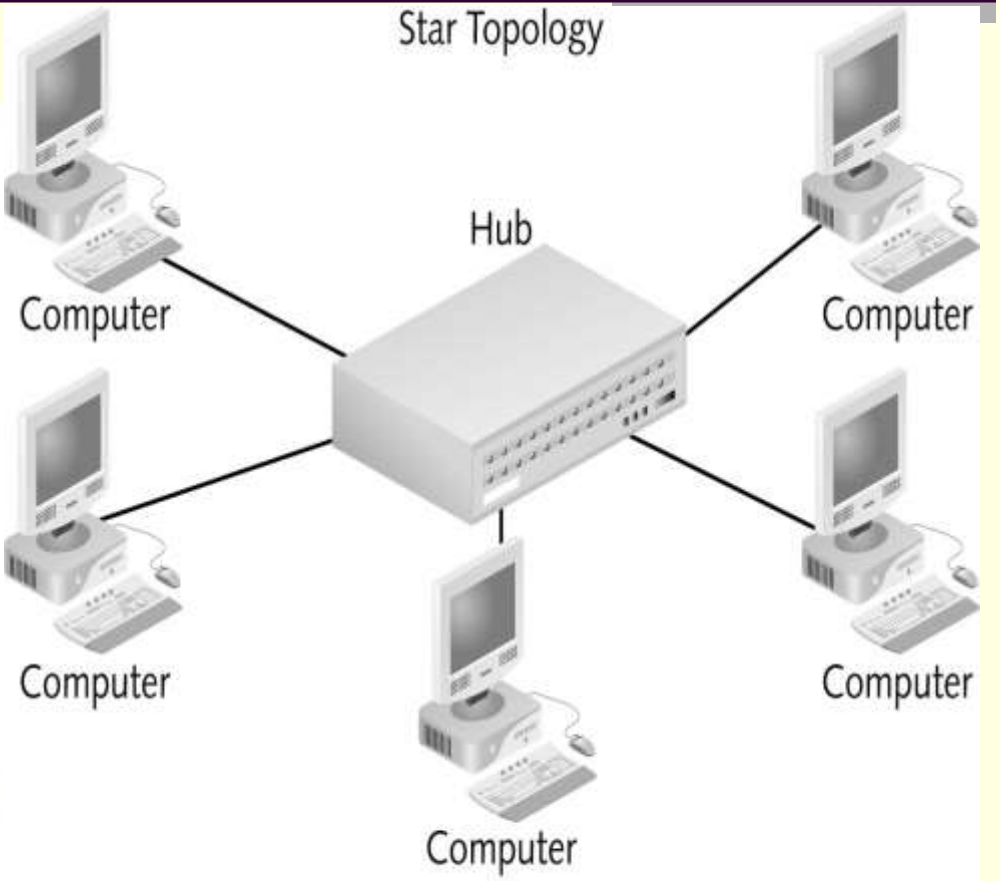


Figure 2-3 Star topology

Common basic networking devices: Bridge

- In telecommunication networks, a **bridge** is a product that connects a local area network (LAN) to another local area network that uses the same protocol (for example, Ethernet or token ring).
- Operate at the Data Link layer

Bridges

A bridge can be used to connect two similar LANs, such as two CSMA/CD LANs.

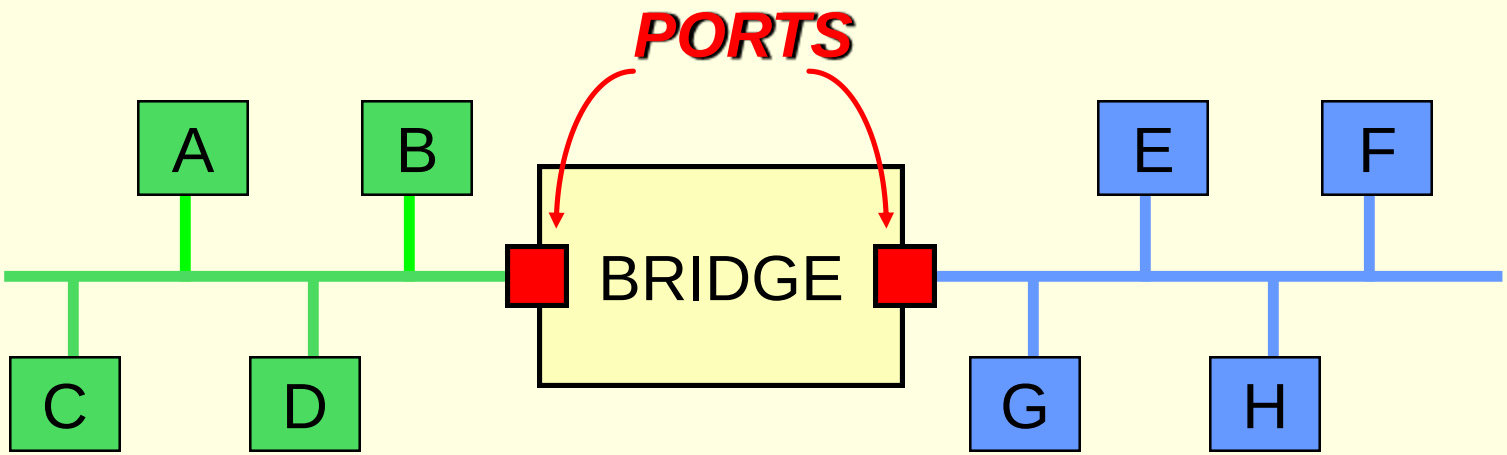
A bridge can also be used to connect two closely similar LANs, such as a CSMA/CD LAN and a token ring LAN.

The bridge examines the destination address in a frame and either forwards this frame onto the next LAN or does not.

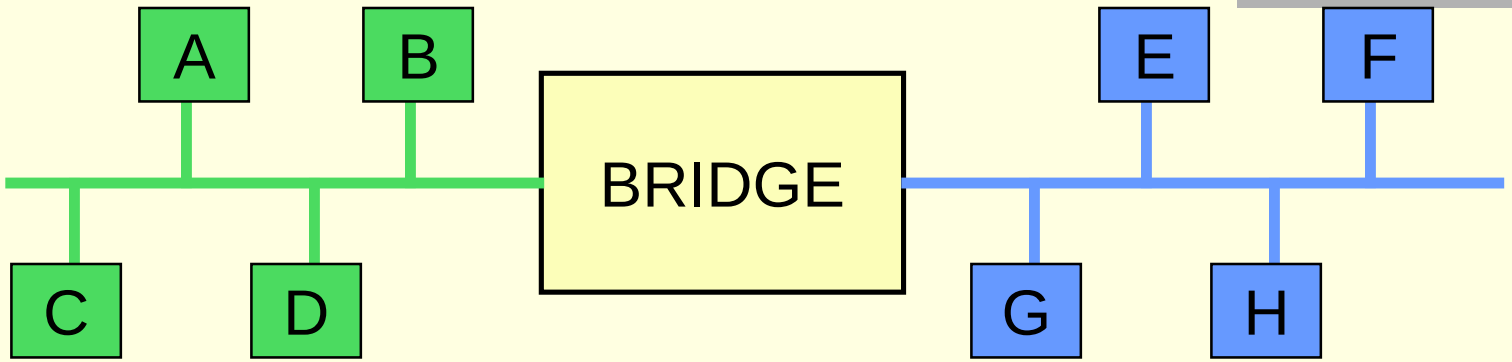
The bridge examines the source address in a frame and places this address in a routing table, to be used for future routing decisions.

Common basic networking devices: Bridge

- A bridge connects networks and forwards frames from one network to another.



Selective Forwarding



- If A sends a frame to E - the frame must be forwarded by the bridge.
- If A sends a frame to B - there is no reason to forward the frame.

Common basic networking devices: Bridge

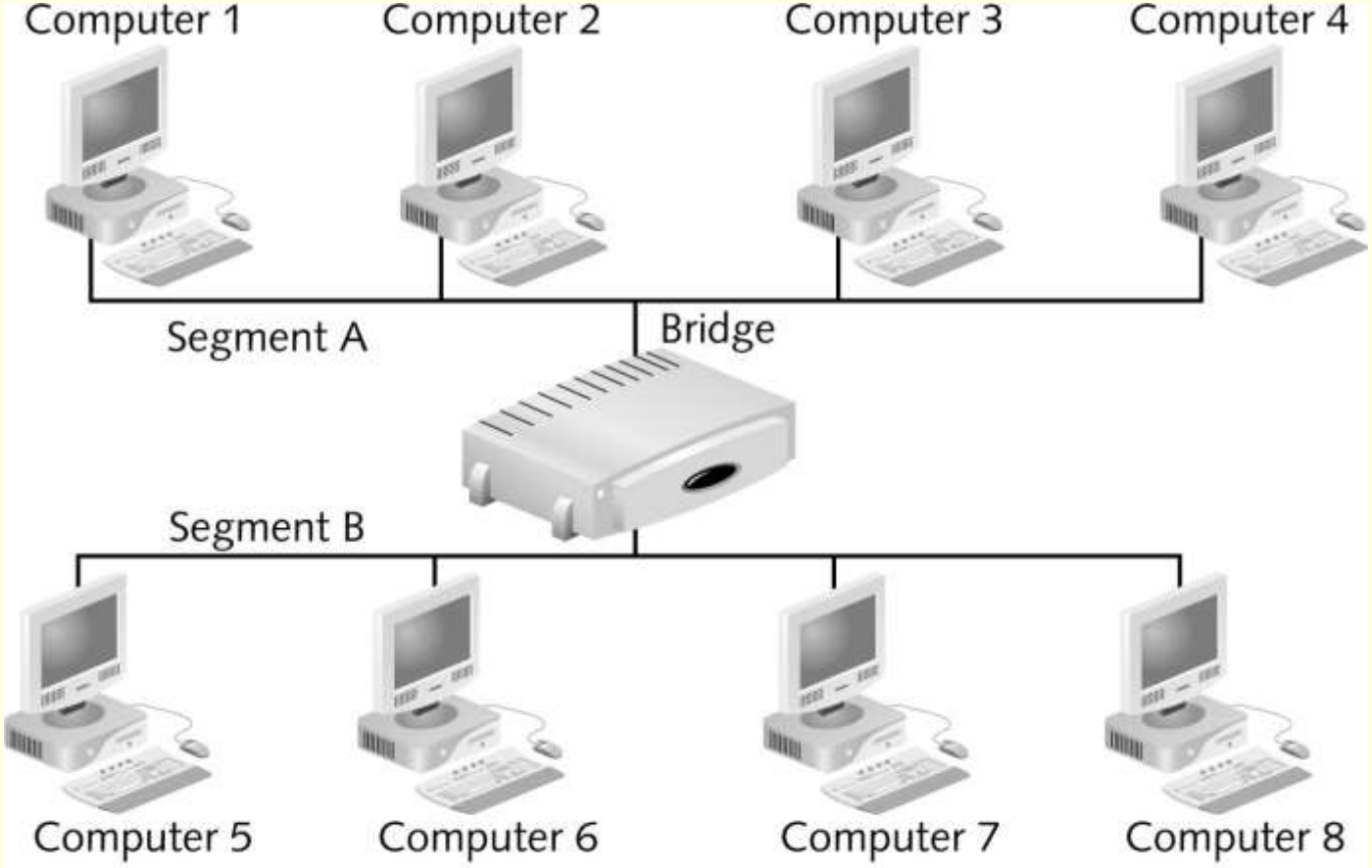
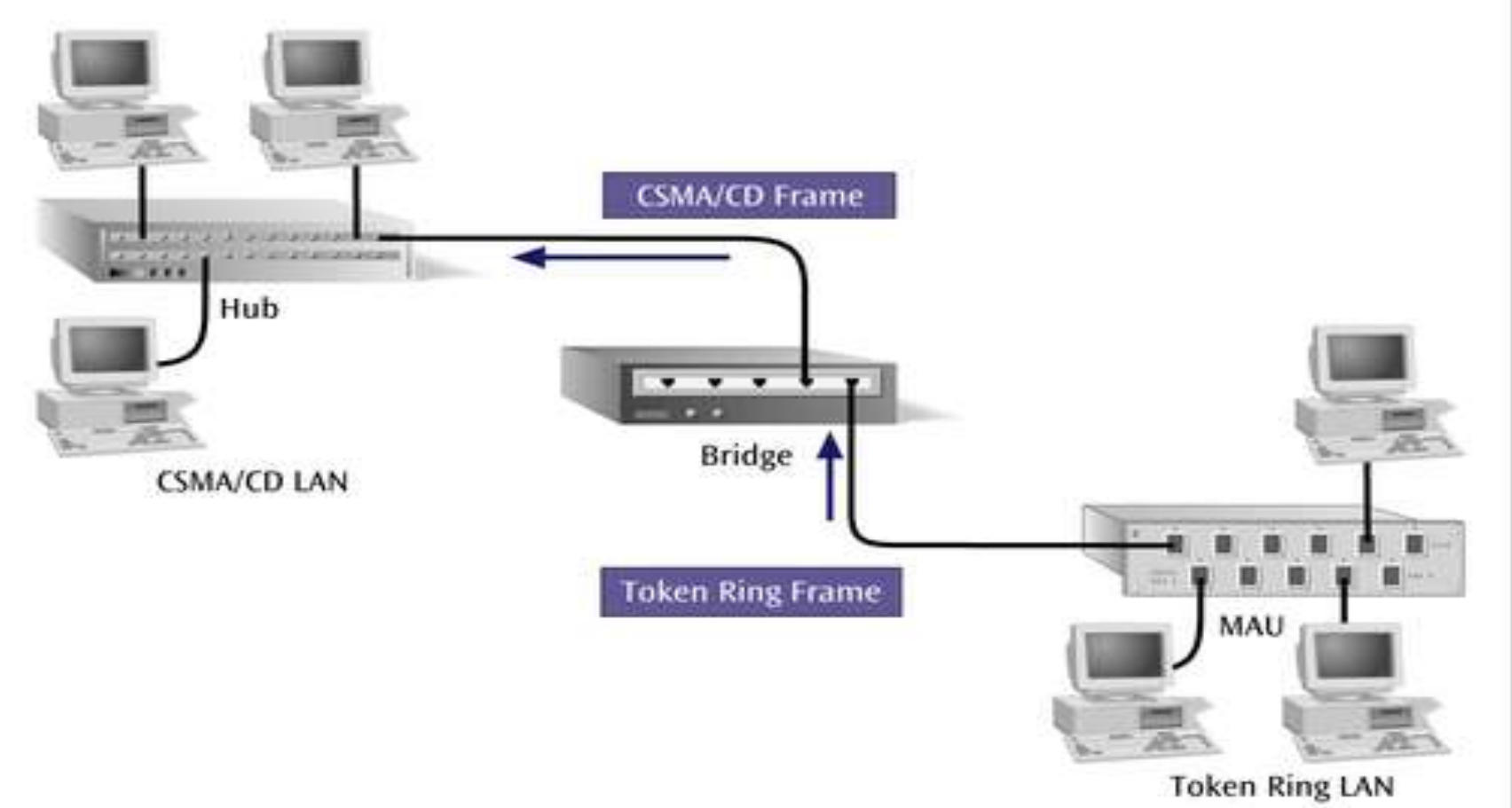
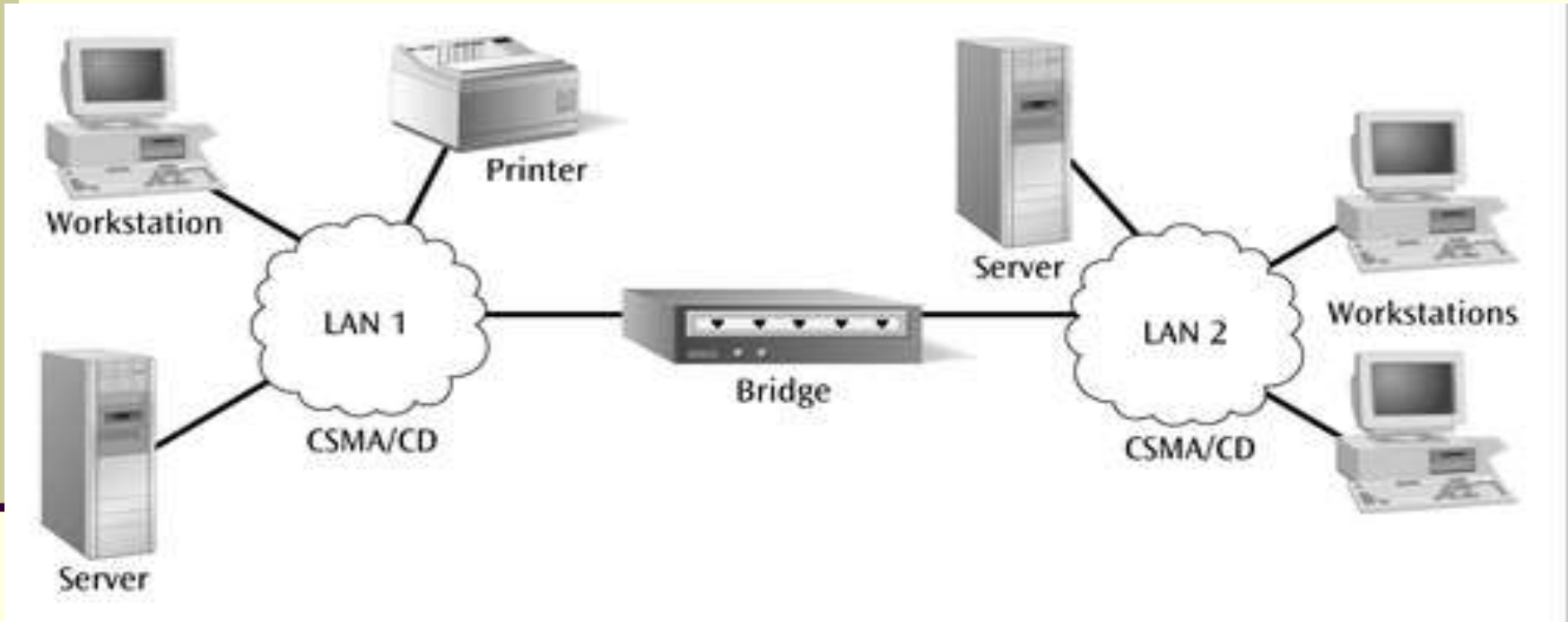


Figure 2-6 Bridge

A bridge interconnecting a CSMA/CD LAN and a token ring LAN.



Bridge interconnecting two identical LANs



Advantages of network bridges

- Extend physical network
- Reduce network traffic with minor segmentation
- Creates separate collision domains
- Reduce collisions
- Connect different architecture

Disadvantages of network bridges

- Does not limit the scope of broadcasts [broadcast domain cannot be controlled]
- Does not scale to extremely large networks
- Buffering and processing introduces delays
- Bridges are more expensive than repeaters or hubs

Common basic networking devices: **Switches**

- A **network switch** is a small hardware device that joins multiple computers together within one local area network (LAN).
- Technically, network switches operate at layer two (Data Link Layer) of the OSI model.

Switches

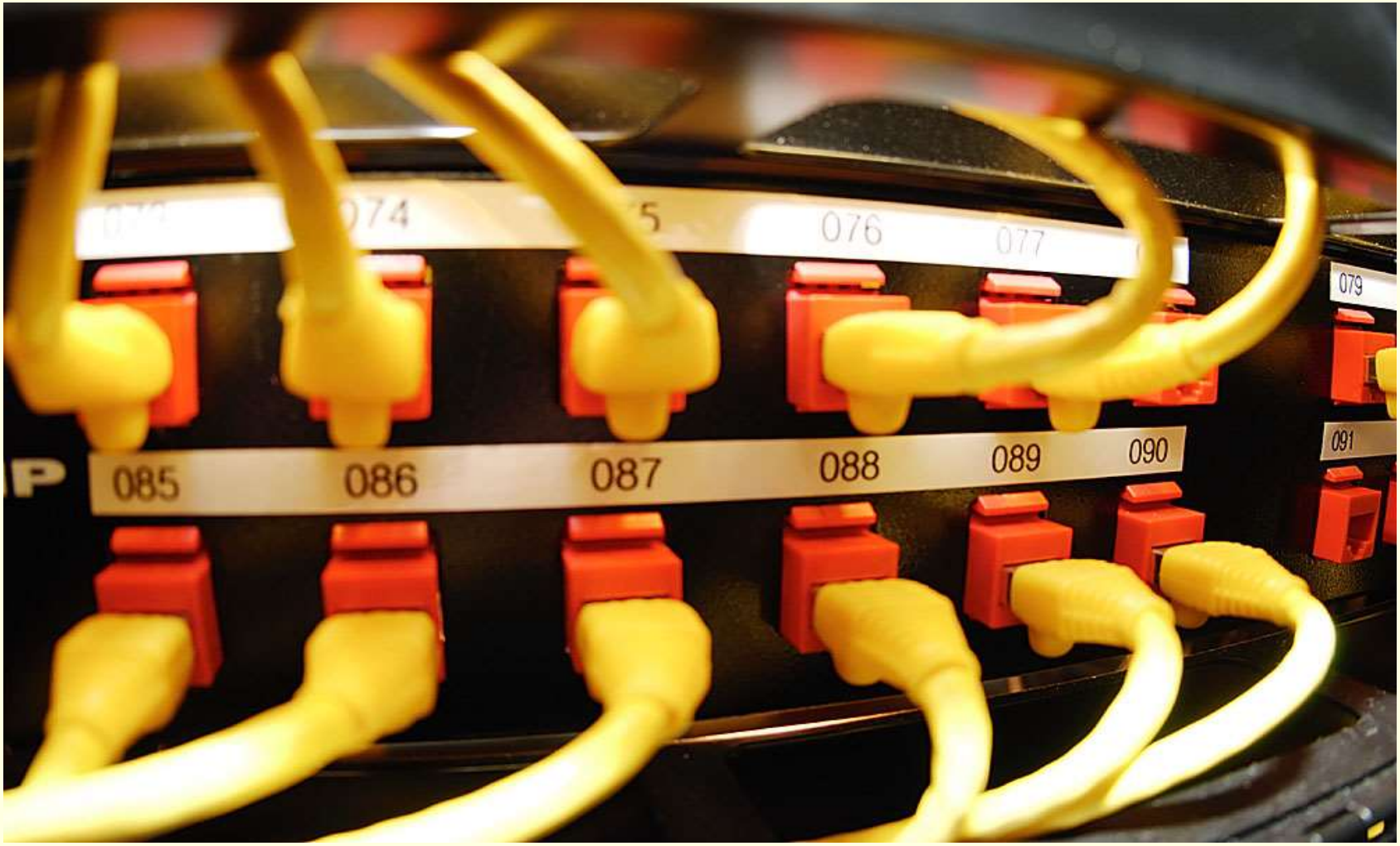
A switch is a combination of a hub and a bridge.

It can interconnect two or more workstations, but like a bridge, it observes traffic flow and learns.

When a frame arrives at a switch, the switch examines the destination address and forwards the frame out the one necessary connection.

- Workstations that connect to a hub are on a *shared segment*.
- Workstations that connect to a switch are on a *switched segment*.

Common basic networking devices: **Switches**



Switches (continued)

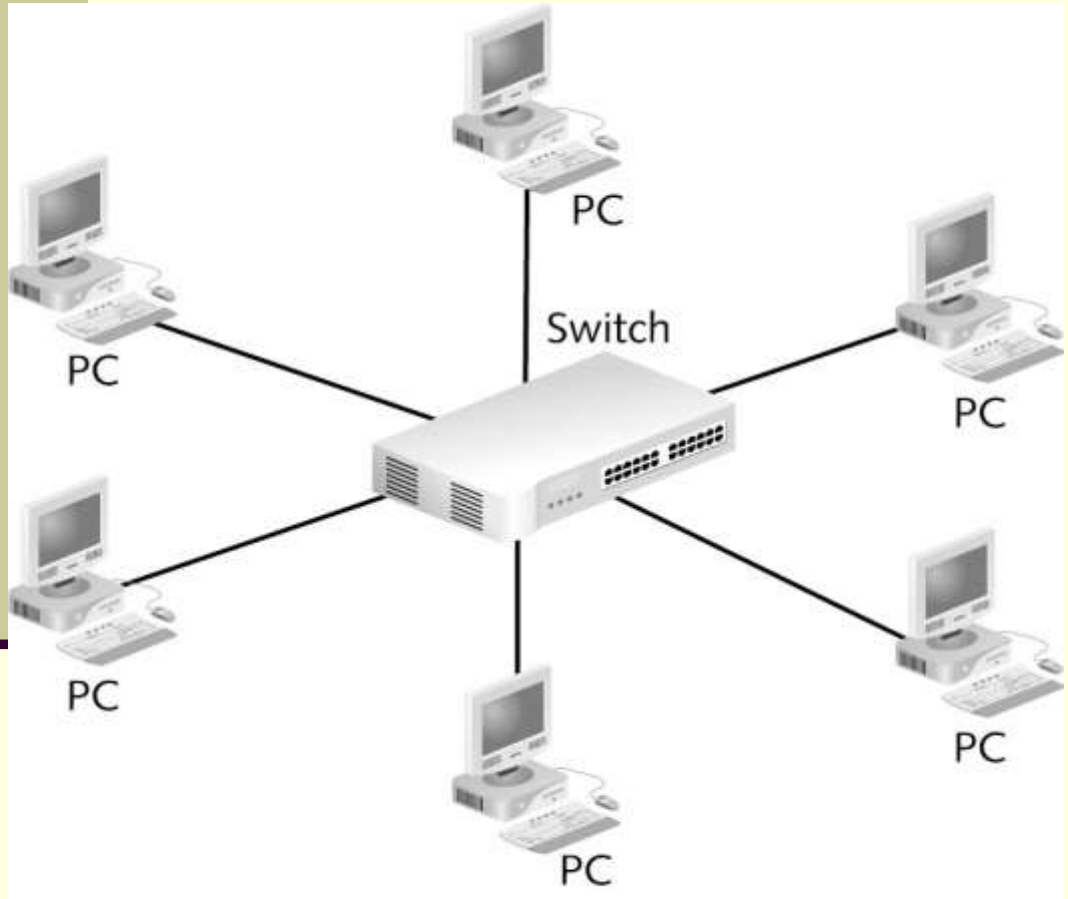
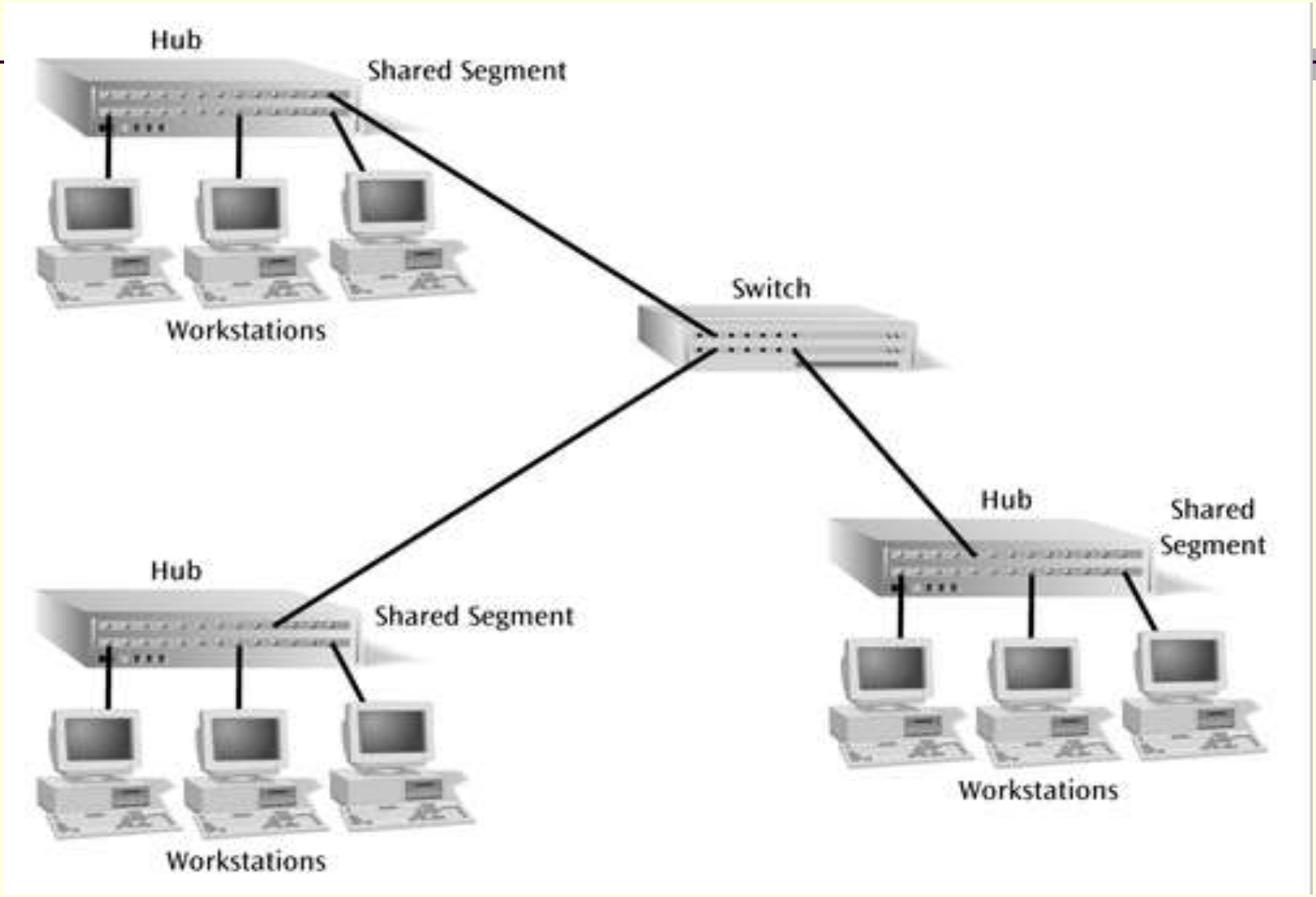
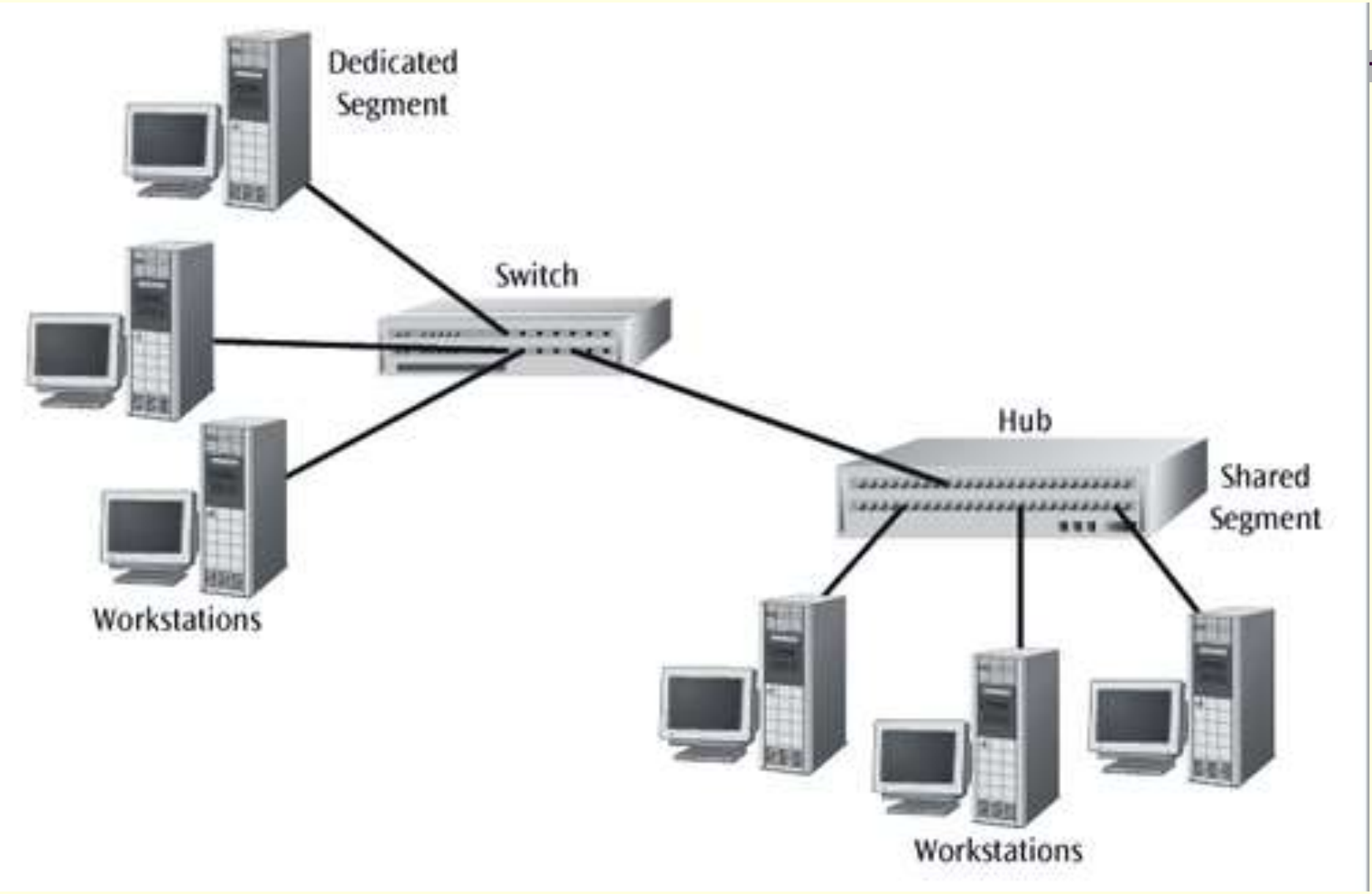


Figure 2-7 Star topology using a switch

Workstations connected to a shared segment of a LAN



Workstations connected to a dedicated segment of a LAN



Advantages of switches

- Increase available network bandwidth
- Reduced workload, computers only receive packets intended for them specifically
- Increase network performance
- Smaller collision domains

Disadvantages of switches

- More expensive than hubs and bridges
- Difficult to trace network connectivity problems through a switch
- Does not filter broadcast traffic

Switch vs Hub

- Hubs and switches are different types of network equipment that connect devices. They differ in the way that they pass on the network traffic that they receive.
- **Hubs**
- The term ‘hub’ is sometimes used to refer to any piece of network equipment that connects PCs together, but it actually refers to a multi-port repeater. This type of device simply passes on (repeats) all the information it receives, so that all devices connected to its ports receive that information.
- Hubs repeat everything they receive and can be used to extend the network. However, this can result in a lot of unnecessary traffic being sent to all devices on the network. Hubs pass on traffic to the network regardless of the intended destination; the PCs to which the packets are sent use the address information in each packet to work out which packets are meant for them. In a small network repeating is not a problem but for a larger, more heavily used network, another piece of networking equipment (such as a switch) may be required to help reduce the amount of unnecessary traffic being generated.
- **Switches**
- Switches control the flow of network traffic based on the address information in each packet. A switch learns which devices are connected to its ports (by monitoring the packets it receives), and then forwards on packets to the appropriate port only. This allows simultaneous communication across the switch, improving bandwidth.
- This switching operation reduces the amount of unnecessary traffic that would have occurred if the same information had been sent from every port (as with a hub).
- Switches and hubs are often used in the same network; the hubs extend the network by providing more ports, and the switches divide the network into smaller, less congested sections.

Switch vs Hub

	Hub	Switch
Technical Specifications:	Hubs classify as Layer 1 devices in the OSI model	Network switches operate at layer two (Data Link Layer) of the OSI model.
Cost:	Cheaper than switches	Costlier than hubs
Manufacturers:	Sun Systems, Oracle and Cisco	Sun Systems, Oracle, Belkin, Linksys, and Net Gear, Huawei
Function:	To connect a network of personal computers together, they can be joined through a central hub	Network switches inspect data packets as they are received, determining the source and destination device of that packet, and forwarding it appropriately
Definition:	An electronic device that connects many computers together to form a single computer network	A way of routing electricity and data flow patterns through circuits based on binary decisions

When Should I Use a Hub or Switch?

- In a small network (less than 30 users), a hub (or collection of hubs) can easily cope with the network traffic generated and is the ideal piece of equipment to use for connecting the users.
- When the network gets larger (about 50 users), you may need to use a switch to divide the groups of hubs, to cut down the amount of unnecessary traffic being generated.

Common basic networking devices: **Router**

- A network router is a network device with interfaces in multiple networks whose task is to copy packets from one network to another.
- Routers operate at Layer 3 of the OSI Model, the Network Layer. This is in contrast to switches, which operate at Layer 2 of the OSI Model, the Data-Link Layer

A Cisco 7200 Series Router

Routers

- Provide filtering and network traffic control
- Used on LANs and WANs
- Connect multiple segments and networks
- Multiple routers create an “internetwork”
- Operate at the Network layer

Routers (continued)

- Create a table to determine how to forward packets
- Filtering and traffic control base on logical addresses

Routers (continued)

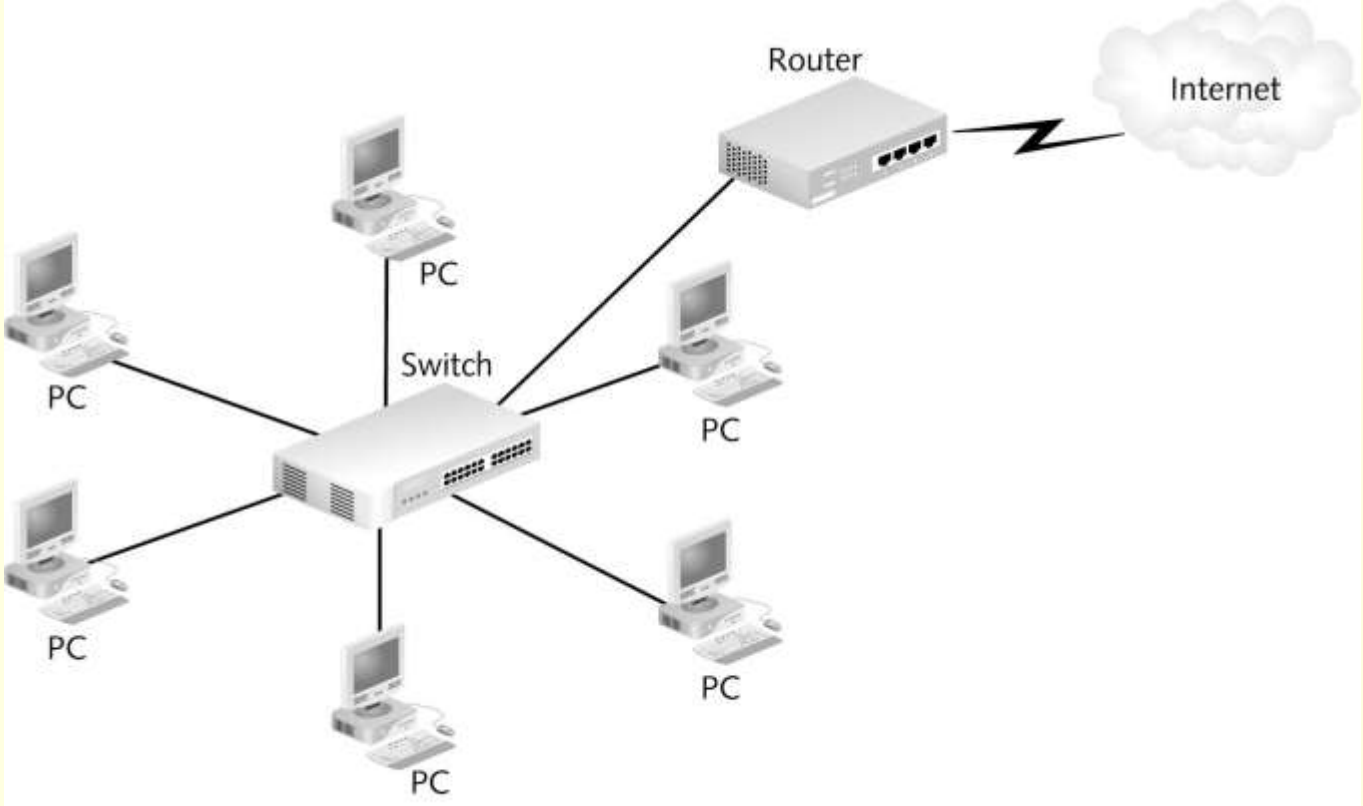
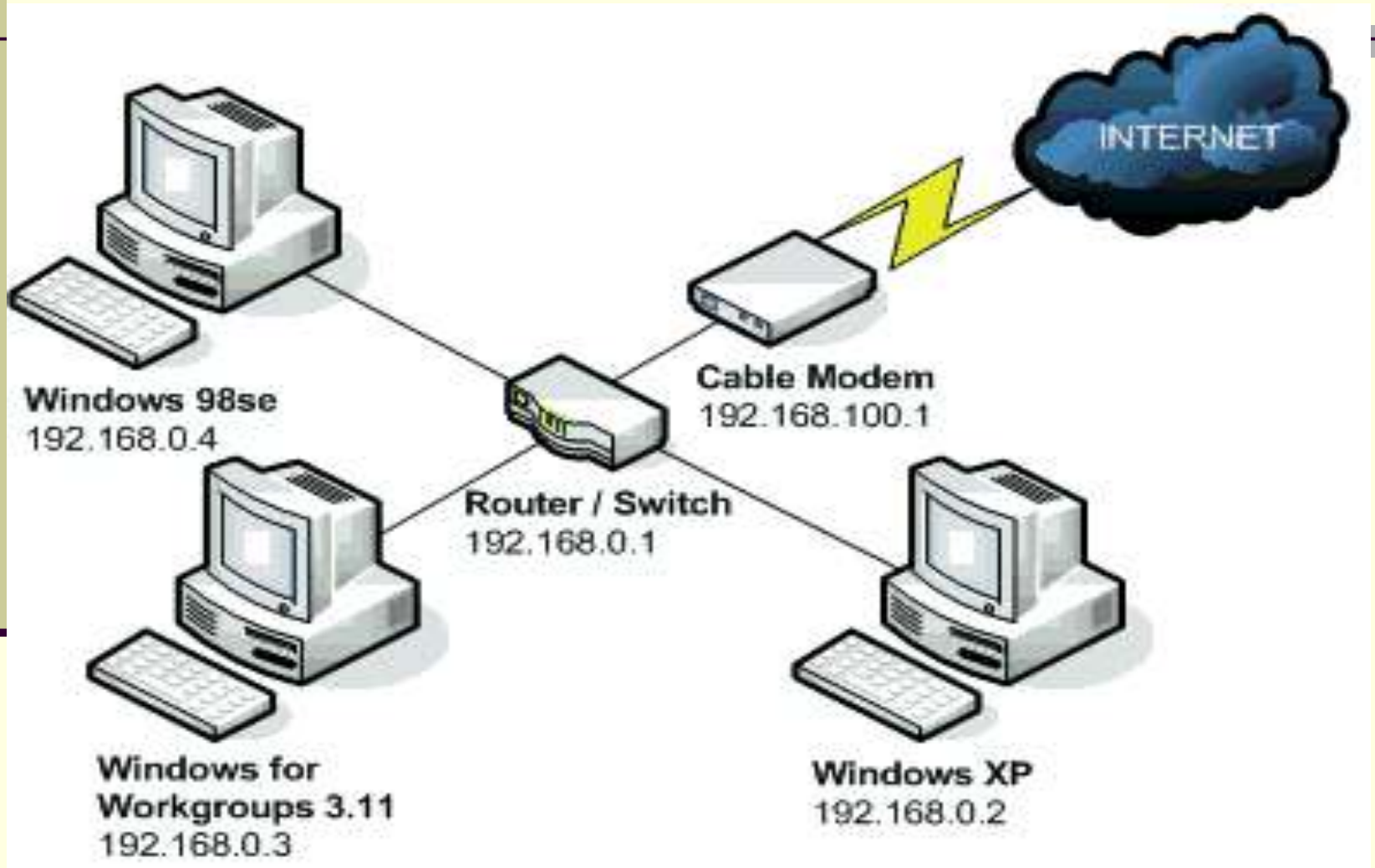
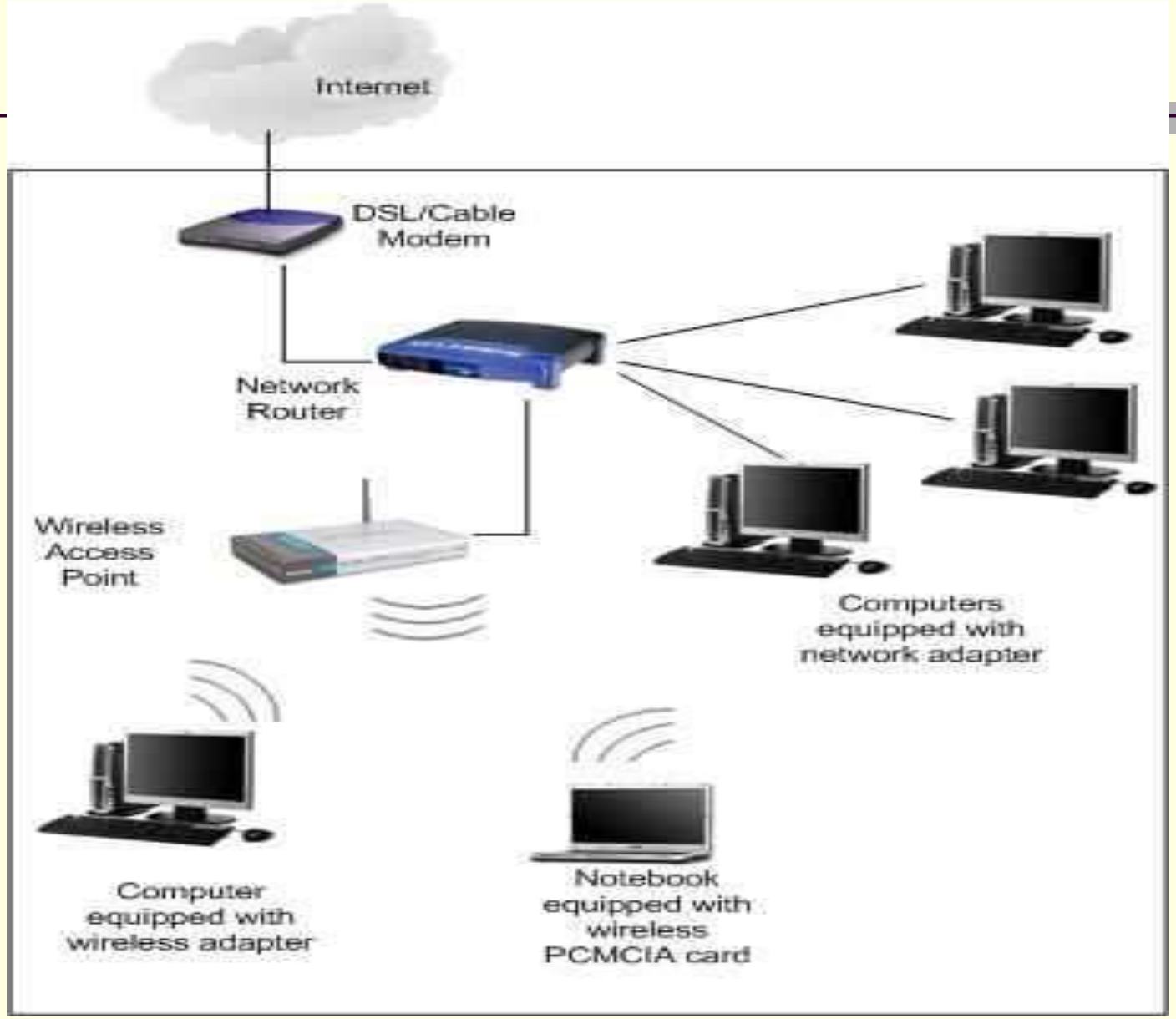


Figure 2-9 Router connecting network to the Internet

Routers (continued)



Routers (continued)



Advantages of routers

- Can connect networks of different architecture
  - Token Ring to Ethernet
- Choose best path through or to a network
- Create smaller collision domains
- Create smaller broadcast domains

Disadvantages of routers

- Only work with routable protocols
- More expensive than hubs, bridges, and switches
- Routing table updates consume bandwidth
- Increase latency due to a greater degree of packet filtering and/or analyzing

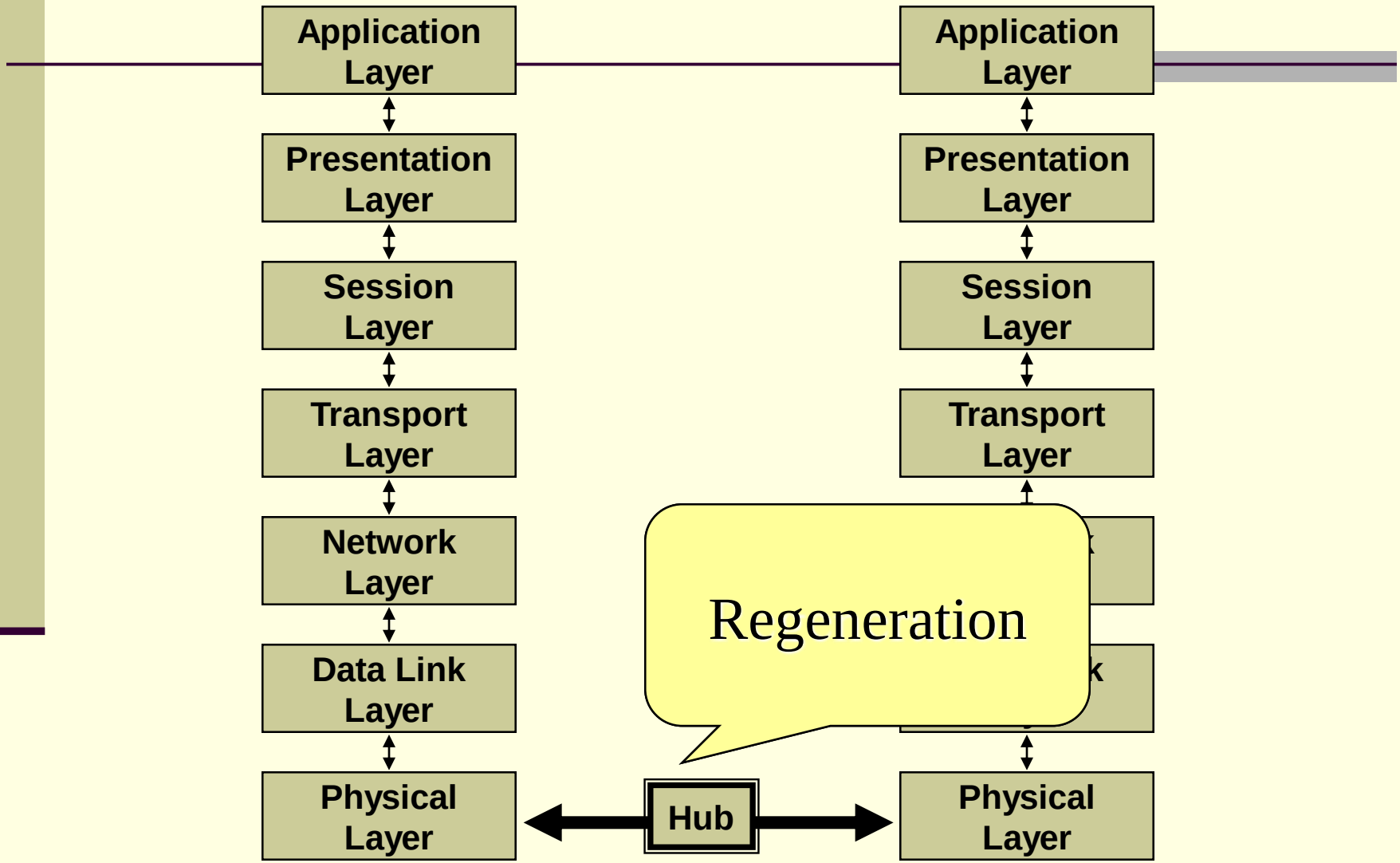
Bridges vs Routers

Bridge: A bridge is a device that connects two segments of the *same* network. The two networks being connected can be alike or dissimilar. Unlike routers, bridges are *protocol-independent*. They simply forward packets without analyzing and re-routing messages.

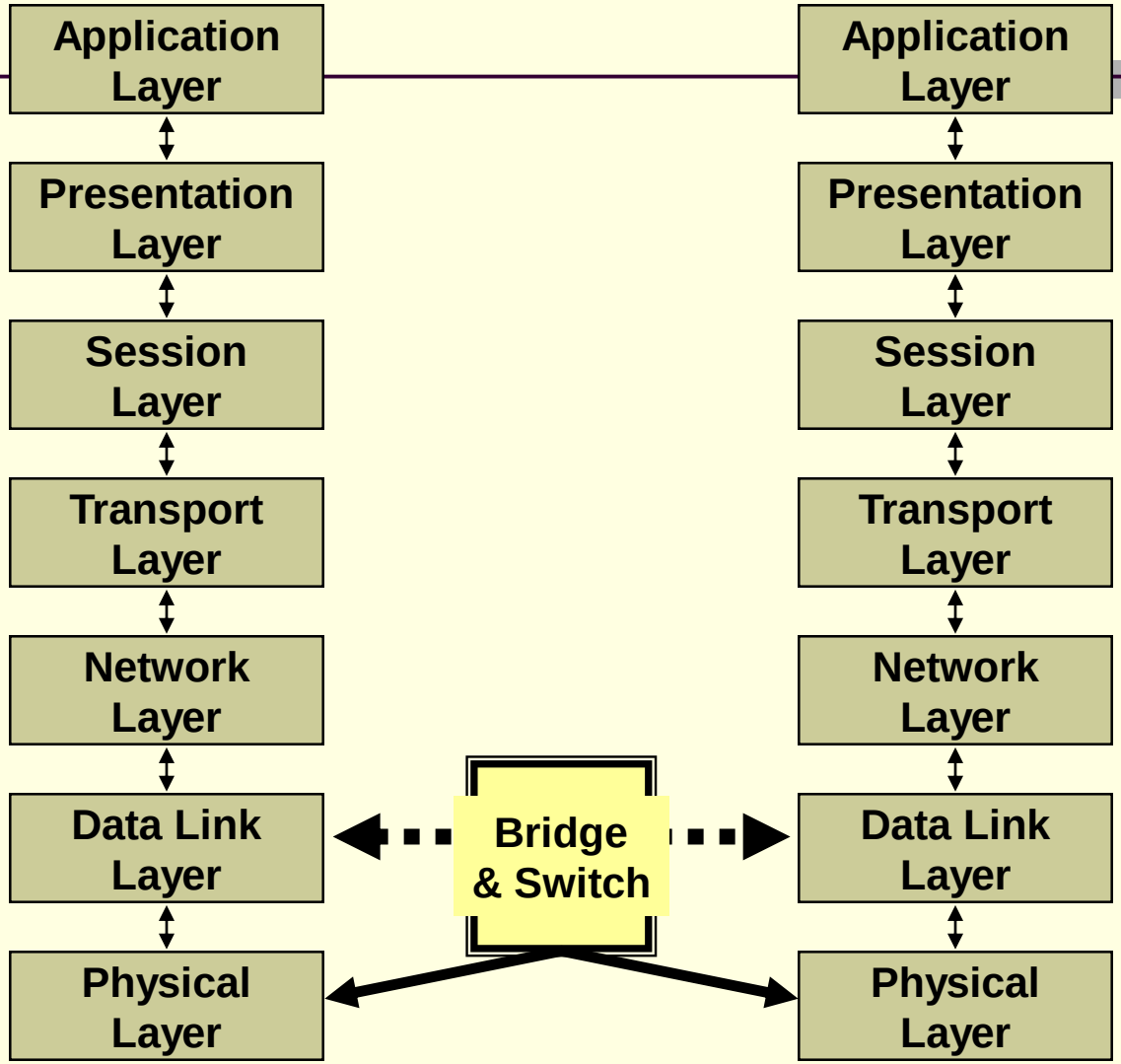
Router: A router is a device that connects two *distinct* networks. Routers are similar to bridges, but provide additional functionality, such as the ability to filter messages and forward them to different places based on various criteria. The Internet uses routers extensively to forward packets from one host to another.

- Based on these definitions we can see that key difference between a bridge and router is that a bridge does not look at protocols and a router does. A bridge does not look at traffic for the purpose of allowing or disallowing it, and it does not decide what to do with certain types of traffic; it simply moves data from one network to another. Whereas a router examines protocols and decides what to do with each packet based on defined criteria.

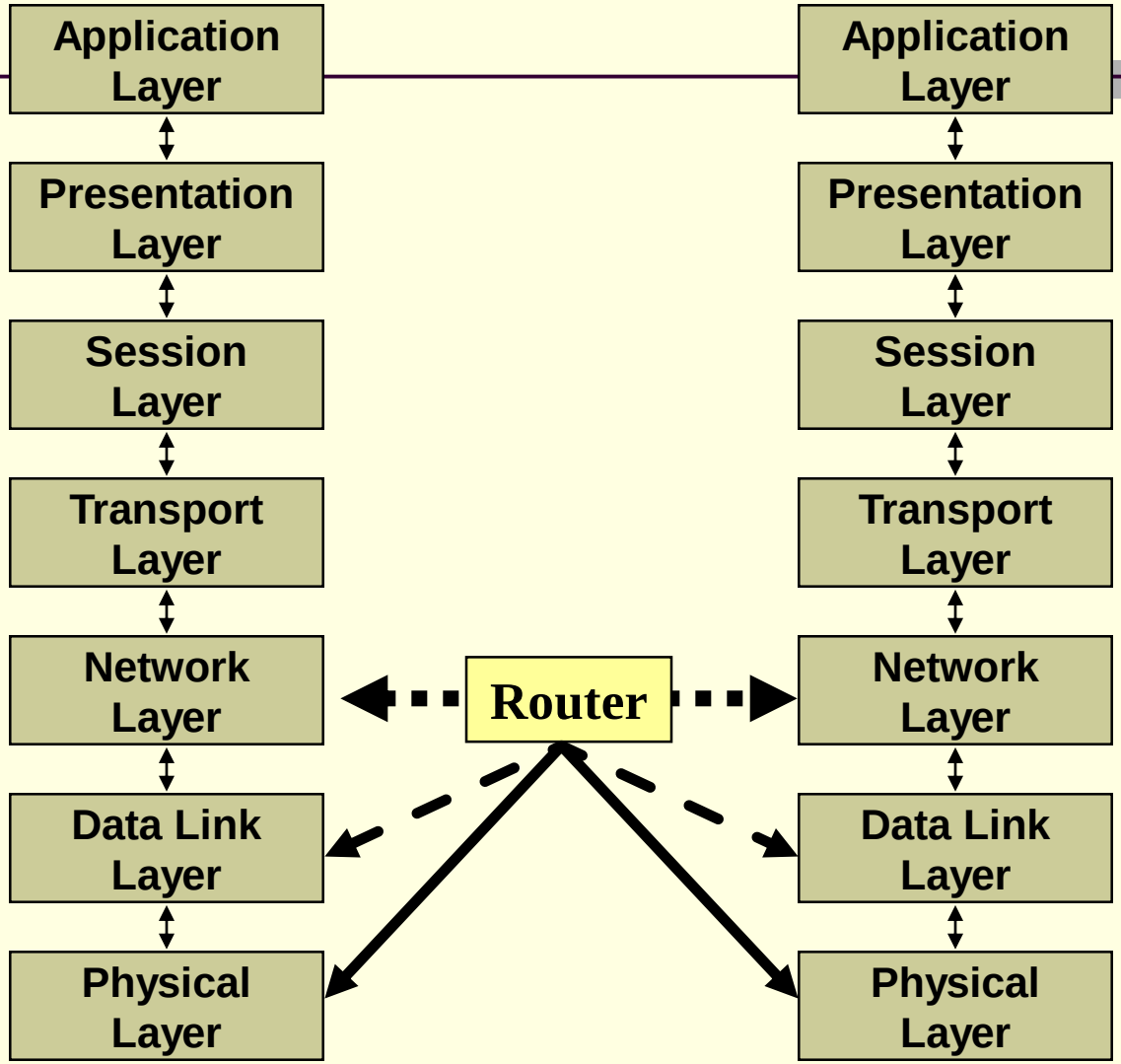
The Physical Layer Connection



The Data Link Connection



The Network Layer Connection



Hybrid network devices :Multilayer Switch

- A **multilayer switch (MLS)** is a computer networking device that switches on OSI layer 2 like an ordinary network switch and provides extra functions on higher OSI layers.

Hybrid network devices :Brouters

- A **Bridge Router** or **brouter** is a network device that works as a bridge and as a router. The brouter routes packets for known protocols and simply forwards all other packets as a bridge would.
- Brouters operate at both the network layer for routable protocols and at the data link layer for non-routable protocols.
- As networks continue to become more complex, a mix of routable and non-routable protocols has led to the need for the combined features of bridges and routers.

Hybrid network devices :Brouters

- Brouters handle both routable and non-routable features by acting as routers for routable protocols and bridges for non-routable protocols.
- Bridged protocols might propagate throughout the network, but techniques such as filtering and learning might be used to reduce potential congestion.
- Brouters are used as connecting devices in the networking system, so it acts as a bridge in a network and as a router in an internetwork.

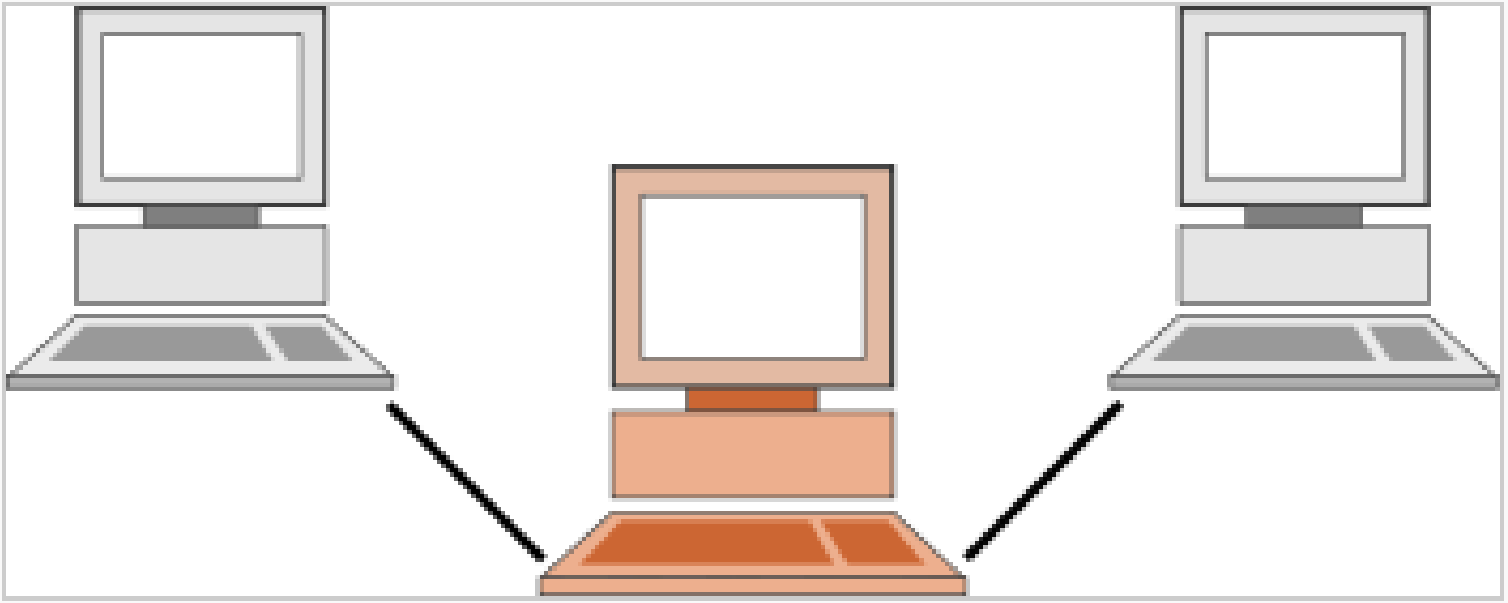
Hybrid network devices:Digital Media Receiver

- A **digital media receiver** (DMR), also commonly referred to as a **media extender**, **media streamer**, or **digital media adapter** (DMA), is a home entertainment device that can connect to a home network to retrieve digital media files (such as music, pictures, or video) from a personal computer or other networked media server and play them back on a home theater system or TV.

Proxy server

- In computer networks, a **proxy server** is a server (a computer system or an application program) that acts as an intermediary for requests from clients seeking resources from other servers.
- A client connects to the proxy server, requesting some service, such as a file, connection, web page, or other resource, available from a different server.
- The proxy server evaluates the request according to its filtering rules. For example, it may filter traffic by IP address or protocol. If the request is validated by the filter, the proxy provides the resource by connecting to the relevant server and requesting the service on behalf of the client.
- A proxy server may optionally alter the client's request or the server's response, and sometimes it may serve the request without contacting the specified server. In this case, it 'caches' responses from the remote server, and returns subsequent requests for the same content directly.

Proxy server



Schematic representation of a proxy server, where the computer in the middle acts as the proxy server between the other two.

Proxy server

- Most proxies are a **web proxy**, allowing access to content on the World Wide Web.
- A proxy server has a large variety of potential purposes, including:
  - To keep machines behind it anonymous (mainly for security).
  - To speed up access to resources (using caching). Web proxies are commonly used to cache web pages from a web server.
  - To apply access policy to network services or content, e.g. to block undesired sites.
  - To log / audit usage, i.e. to provide company employee Internet usage reporting.
  - To bypass security/ parental controls.
  - To scan transmitted content for malware before delivery.
  - To scan outbound content, e.g., for data leak protection.
  - To circumvent regional restrictions

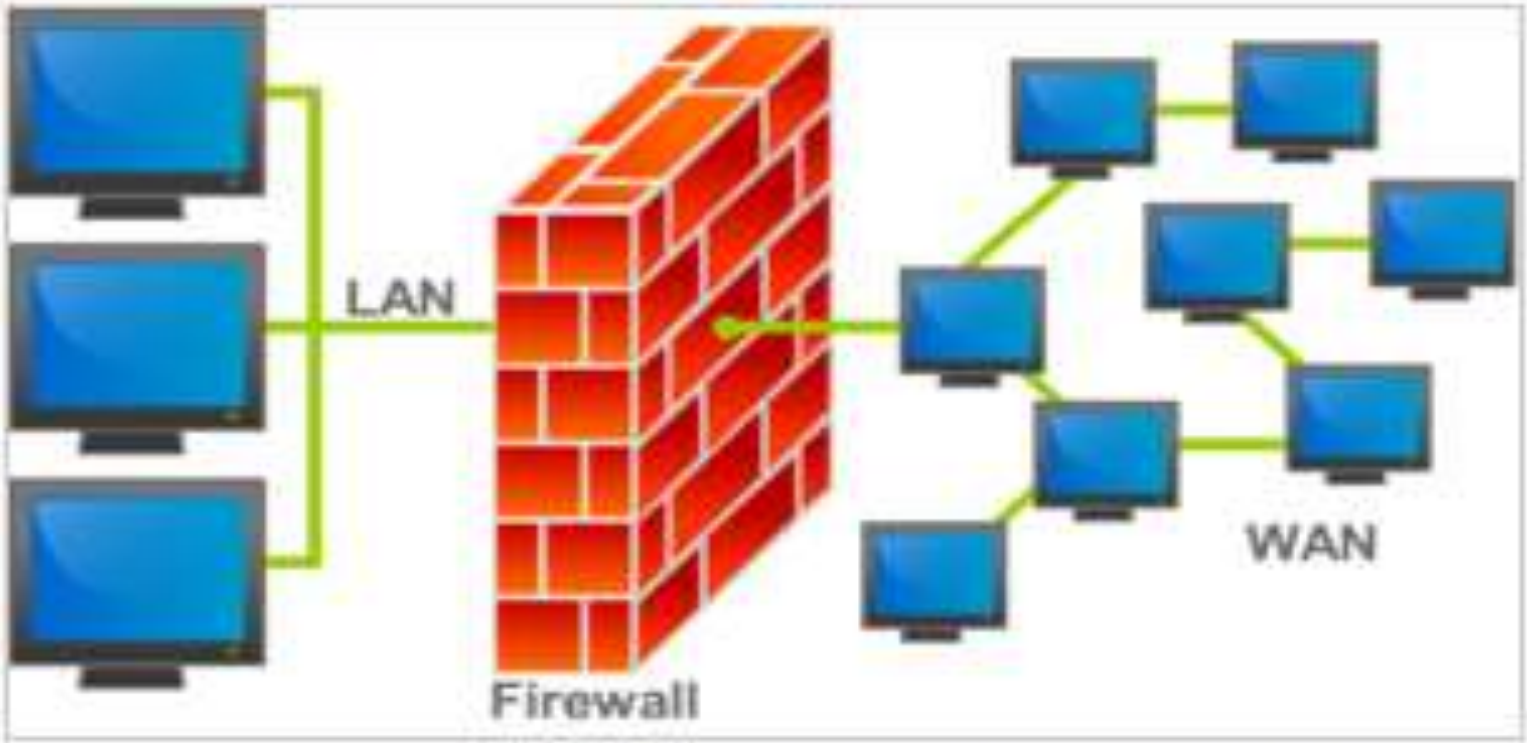
Proxy server

- A proxy server that passes requests and replies unmodified is usually called a gateway or sometimes *tunneling proxy*.
- A proxy server can be placed in the user's local computer or at various points between the user and the destination servers on the Internet.
- A reverse proxy is (usually) an Internet-facing proxy used as a front-end to control and protect access to a server on a private network, commonly also performing tasks such as load-balancing, authentication, decryption or caching.

Firewall

- A **firewall** is a part of a computer system or network that is designed to block unauthorized access while permitting authorized communications. It is a device or set of devices that is configured to permit or deny network transmissions based upon a set of rules and other criteria.
- Firewalls can be implemented in either hardware or software, or a combination of both. Firewalls are frequently used to prevent unauthorized Internet users from accessing private networks connected to the Internet, especially intranets.
- All messages entering or leaving the intranet pass through the firewall, which inspects each message and blocks those that do not meet the specified security criteria.

Firewall



An illustration of where a firewall would be located in a network.

Firewall

- There are several types of firewall techniques:
- Packet filter: Packet filtering inspects each packet passing through the network and accepts or rejects it based on user-defined rules. Although difficult to configure, it is fairly effective and mostly transparent to its users.
- Application gateway: Applies security mechanisms to specific applications, such as FTP and Telnet servers. This is very effective, but can impose a performance degradation.
- Circuit-level gateway: Applies security mechanisms when a TCP or UDP connection is established. Once the connection has been made, packets can flow between the hosts without further checking.
- Proxy server: Intercepts all messages entering and leaving the network. The proxy server effectively hides the true network addresses.

Network address translation

- In computer networking, **network address translation** (**NAT**) is the process of modifying network address information in datagram (IP) packet headers while in transit across a traffic routing device for the purpose of remapping one IP address space into another.

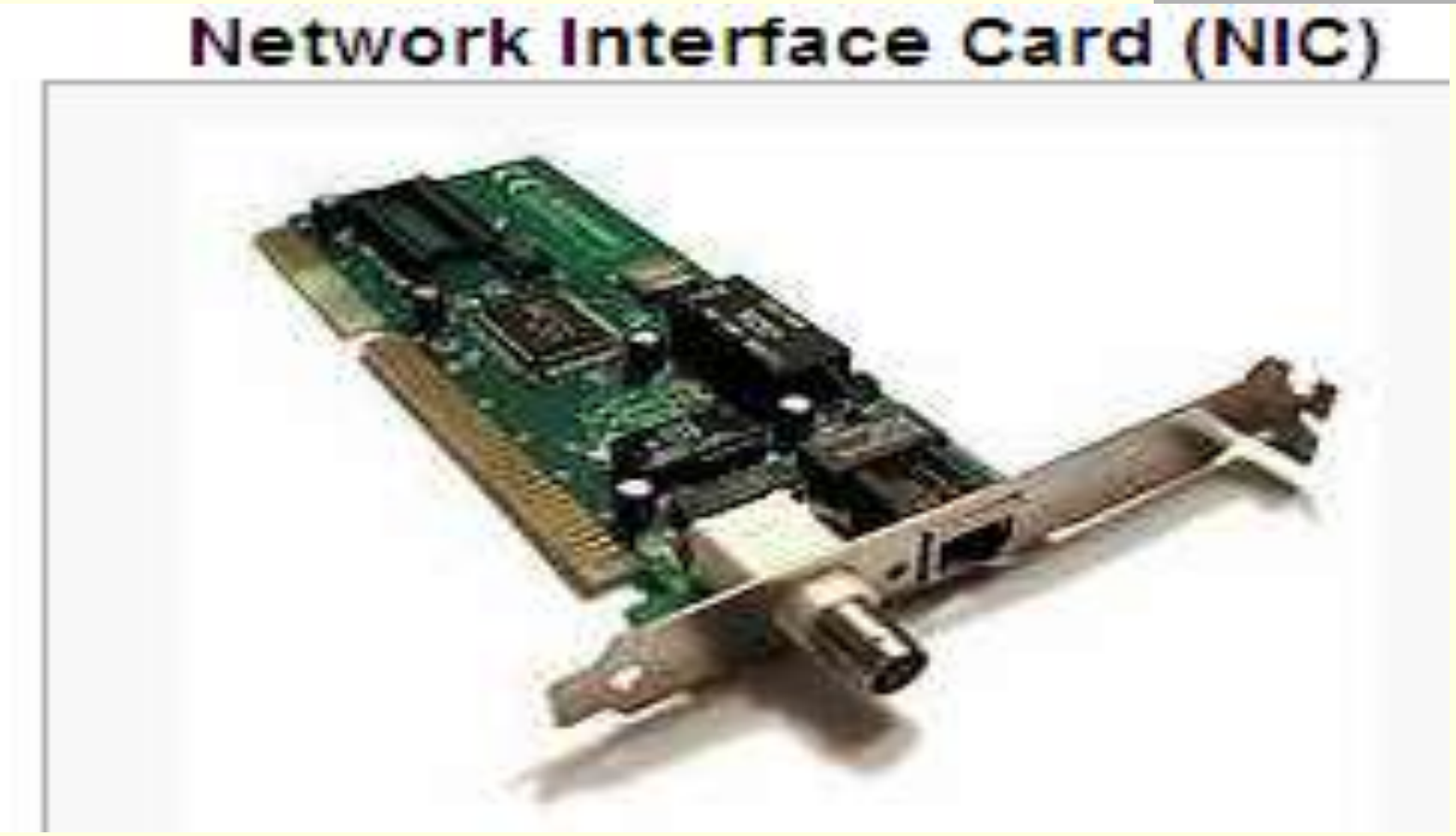
Multiplexer

- In telecommunications, a multiplexer is a device that combines several input information signals into one output signal, which carries several communication channels, by means of some multiplex technique.
- A **demultiplexer** is in this context a device taking a single input signal that carries many channels and separates those over multiple output signals.

Network Card

- A **network interface controller** is a computer hardware component that connects a computer to a computer network. The controller may also be referred to as a **network interface card**, **network adapter**, or a **LAN adapter**.
- Whereas network interface controllers were commonly implemented on expansion cards that plug into a computer bus and known as a **network interface card**, **network card** or **LAN card**, the low cost and ubiquity of the Ethernet standard means that most newer computers have a network interface built into the motherboard.

Network Card



Modem

- A **modem** (**mo**dulator-**dem**odulator) is a device that modulates an analog carrier signal to encode digital information, and also demodulates such a carrier signal to decode the transmitted information. The goal is to produce a signal that can be transmitted easily and decoded to reproduce the original digital data. Modems can be used over any means of transmitting analog signals, from driven diodes to radio.

Broadband Internet access

- **Broadband Internet access**, often shortened to just "broadband", is a high data rate connection to the internet .
- Broadband is often called "**high-speed**" access to the Internet, because it usually has a high rate of data transmission.
- In general, any connection to the customer of 256 kbit/s (0.25 Mbit/s) or greater is more concisely considered **broadband Internet access**.
- The International Telecommunication Union Standardization Sector (ITU-T) recommendation I.113 has defined broadband as a transmission capacity that is faster than primary rate ISDN, at 1.5 to 2 Mbit/s.

Broadband Internet access

- Broadband Internet access--commonly referred to as "broadband"--is high speed Internet access.
- Common types of broadband Internet access include Digital Subscriber Line (DSL), which runs via copper telephone lines; cable Internet, which runs on coaxial or fiber cables that are used primarily for television signal; and satellite Internet, which utilizes orbiting satellites to broadcast data into remote regions.

Dial-up Internet access

- **Dial-up Internet access** is a form of Internet access that uses the facilities of the public switched telephone network (PSTN) to establish a dialed connection to an Internet service provider (ISP) via telephone lines.
- The user's computer or router uses an attached modem to encode and decode Internet Protocol packets and control information into and from analogue audio frequency signals, respectively.

Comparison: Speed

- Broadband Internet access speeds vary greatly between technologies, providers and line conditions. In the United States, the Federal Communications Commission (FCC) defines broadband speeds as over 768 kilobits per second (kbps).

- Dial up is much slower than broadband with a top speed of 56 kilobytes per second (kbps). As of 2010, many dial-up service providers add an accelerator to their dial-up service which artificially accelerates the line speed by compressing the data.

Comparison: Availability

- Due to technology constraints, broadband is not available in all areas. Remote areas such as mountains or rural areas often do not have the telecommunications framework to offer terrestrial broadband such as cable or DSL, though satellite service may be available.

  In contrast, dial-up Internet access is available wherever there is an analog phone. Toll charges may apply when calling a long distance access number.

Comparison: Cost

- As of 2010, dial-up Internet access is generally much cheaper than broadband access, starting at $9.99 per month. Some ISPs offer ad supported dial up for free.

  Broadband prices can vary depending on type and provider but entry level cable or DSL service start at $19.99 a month for either service.